

The KKT conditions for [Lagrange Dual Problem] are

1. $f_i(x^*) \leq 0$
2. $h_i(x^*) = 0$
3. $\lambda_i^* \geq 0$
4. $\lambda_i^* f_i(x^*) = 0$
5. $\nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) + \sum_{j=1}^p \nu_j^* \nabla h_j(x^*) = 0$